Práctica tres: Sistema musculoesquelético

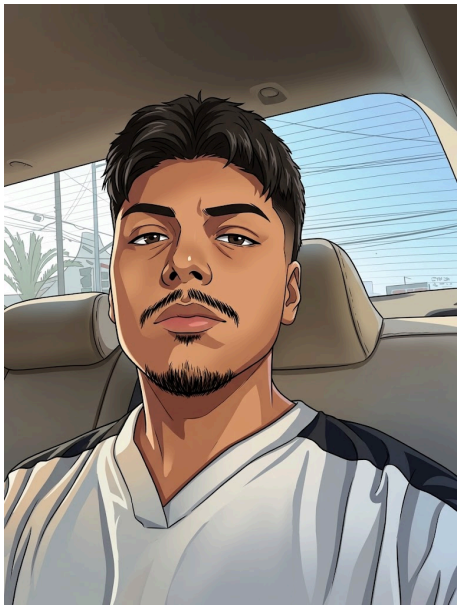
Departamento de Ingeniería Eléctrica y Electrónica, Ingeniería Biomédica

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Información general



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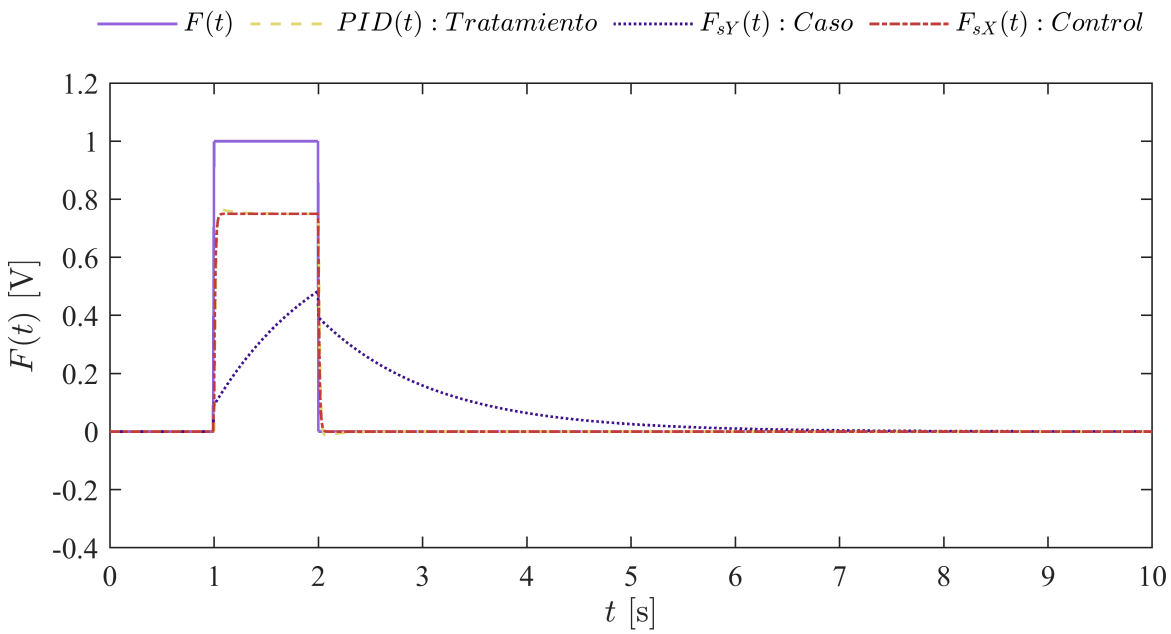
Datos de la simulación

```
clc; clear; close all; warning('off','all')
```

```
clc; clear; close all; warning('off','all')
open_system("practica_3.slx")
parameters.Stoptime='15';
parameters.Solver='ode15s';
parameters.Maxstep='1e-2';
x = sim('practica_3',parameters);
```

Respuesta Control,Caso y Tratamiento

```
plotsignals(x.t, x.F, x.PID, x.Fsy, x.Fsx, 'Respuestas')
```



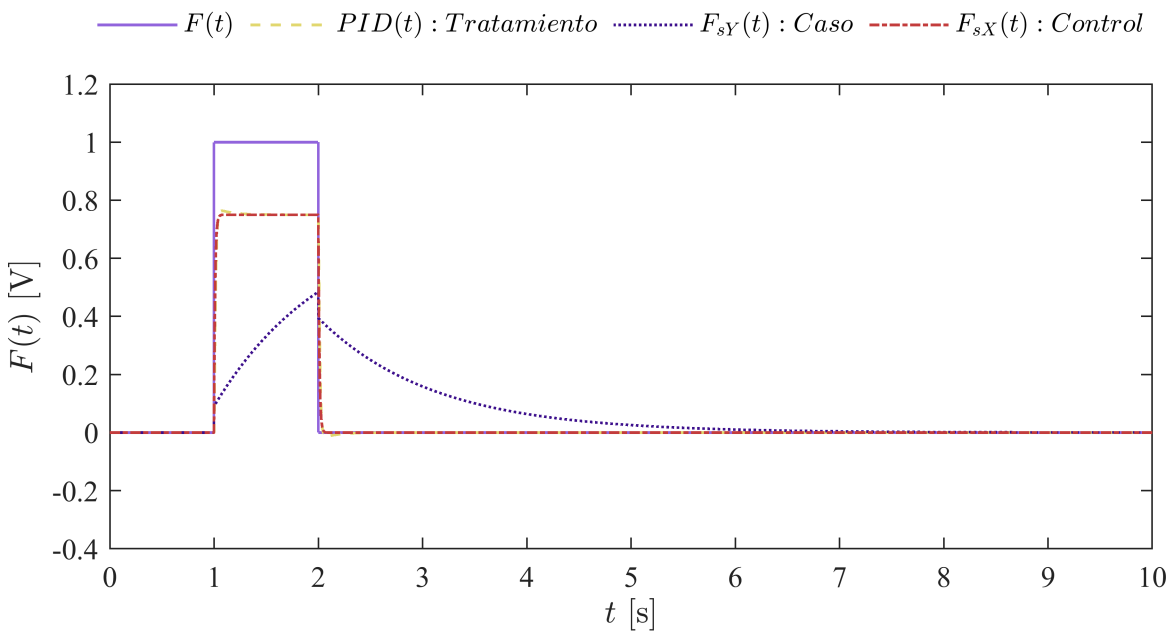
Respuestas EDOS

```
dt = 1e-3;
t= (0:dt:10)';
F = zeros(length(t),1); F(1/dt:2/dt) =1;
PID = interp1(x.t,x.PID,t,'previous');

par_control = [0.25,10e-6,100e-6,100];
par_caso = [0.25,10e-6,100e-6,10e3];

Fs_control = musculoesqueletico(par_control,F,t);
Fs_caso = musculoesqueletico(par_caso,F,t);

plotsignals(t,F,PID,Fs_caso,Fs_control,'EDOsys')
```




```
function Fs = musculoesqueletico(par,F,t)
    alpha = par(1); Cs = par(2); Cp = par(3); R = par(4); dt = t(2);
    n =round(t(end)/dt);
    x = zeros(length(t),1); x(1) = 0;
    Fs = zeros(length(t),1); Fs(1) = 0;
    for i = 1:n
        x(i+1) = x(i) + ((F(i)-x(i))/(R*(Cp+Cs)))*dt; %EDO
        Fs(i+1) = Cs*F(i)/(Cp+Cs) + (1 - alpha -Cs/(Cp+Cs))*x(i); %Funcion
    end
end
```

Funcion: Respuestas de la simulacion

```
function plotsignals (t,x1,x2,x3,x4,signal)
    set(figure(),'color','w')
    set(gcf,'units','inches','position',[1,1,7,3.5])
    set(gca,'fontname','times new roman','fontsize',11)
    hold on; grid off; box on;
    clr1 = [149, 100, 221]/256;
    clr2 = [228, 218, 114]/256;
    clr3 = [62, 15, 141]/256;
    clr4 = [200, 60, 60]/256;
    colororder([clr1;clr2;clr3;clr4])

    if signal == "Respuestas"
        plot(t,x1,'-', t,x2,'--', t,x3,':', t,x4,'-.','LineWidth',1)
        L = legend('$F(t)$','$PID(t): Tratamiento$','$F_{sY}(t): Caso$','$F_{sX}(t): Control$')

    elseif signal == "EDOsys"
```

```
        plot(t,x1,'-', t,x2,'--', t,x3,':', t,x4,'-.','LineWidth',1)
        L = legend('$F(t)$','$PID(t): Tratamiento$','$F_{sY}(t): Caso$','$F_{sX}(t): Control$')
    end

    set(L,'interpreter','latex','location','northoutside','orientation','horizontal','box','off

    xlabel('$t$ [s]','Interpreter','latex')
    ylabel('$F(t)$ [V]','Interpreter','latex')

    xlim([0 10]); xticks(0:1:10)
    ylim([-0.4 1.2]); yticks(-0.4:0.2:1.2)

    exportgraphics(gcf,[signal,'_matlab.pdf'],'ContentType','vector')
end
```